

Zecpath AI

Day 20 — Final ATS Evaluation Report

Production Readiness, Demo Dataset & End-to-End ATS Validation

1. Executive Summary

The Zecpath AI ATS system integrates resume processing, skill extraction, semantic matching, ATS scoring, candidate ranking, and automated shortlisting into an end-to-end recruitment workflow. Day 20 validates the complete ATS module using the components implemented during the internship.

2. End-to-End ATS Workflow

Resume → Text Extraction → Resume Parsing → Skill / Experience / Education Extraction → Semantic Resume-Job Matching → ATS Score → Candidate Ranking → Shortlisting API → Recruitment Decision

3. ATS Scoring Methodology

The scoring engine evaluates Skill Match, Experience Relevance, Education Alignment, and Semantic Similarity using role-specific weights from config/weights.json. Each component is validated between 0 and 100; missing values are handled as zero and the final score is rounded to two decimal places.

Component	Purpose
Skill Match	Alignment between candidate skills and job requirements.
Experience Relevance	Relevance of candidate experience to the target role.
Education Alignment	Alignment of education with role requirements.
Semantic Similarity	Semantic similarity between resume and job content.

4. Accuracy Evaluation

Metric	Result
Resume samples tested	3
Skills successfully detected	17
Missed skills	0
False positives	0
Technology Stack Detection	Successful
Synonym Detection	Successful
Overall Accuracy	100%
Evaluation date	01-Aug-2026

The 100% result reflects the tested three-resume evaluation dataset and should not be interpreted as a universal production accuracy guarantee.

5. Candidate Ranking & Demo Dataset

Rank	Candidate	ATS Score	Decision
1	Alice	95%	Shortlisted
2	John	88%	Shortlisted
3	Sarah	84%	Shortlisted
4	David	76%	Shortlisted
5	Michael	68%	Review

Top candidate: Alice — 95%. Candidates are sorted from highest to lowest ATS score before the recruitment decision is assigned.

6. Shortlisting Decision Rules

ATS Score	Decision
70–100	Shortlisted
50–69	Review
0–49	Rejected

The candidate ranking logic was refined so its decision thresholds match the FastAPI shortlisting endpoint. The API also validates candidate ID and ATS score boundaries.

7. API Validation

Scenario	ATS Score	Expected Status
Review scenario	60	Review
Rejected scenario	40	Rejected

API response fields: candidate_id, ats_score, status, and message.

8. Production Readiness Assessment

Capability	Status
Resume processing	Implemented
Skill extraction	Implemented
ATS scoring	Implemented
Semantic matching	Implemented
Candidate ranking	Implemented

Automated shortlisting	Implemented
API integration	Implemented
Accuracy evaluation	Validated
Decision threshold consistency	Refined

9. Limitations

The reported skill-extraction accuracy is based on a small test sample of three resumes. The 100% result demonstrates performance on that evaluated sample, not a universal production accuracy guarantee. Broader validation would require larger and more diverse datasets across roles, formats, and candidate populations.

The demo ranking dataset contains five candidates and is intended to demonstrate the ranking and decision workflow. Larger datasets would be appropriate for extensive benchmarking.

10. Final Conclusion

Day 20 completes the validation stage of the Zecpath AI ATS system. The project demonstrates an end-to-end recruitment workflow from resume processing through ATS evaluation, candidate ranking, API-based shortlisting, and recruitment decision support.

Day 20 Deliverables

- ✓ Production-ready ATS AI system
- ✓ Demo datasets
- ✓ Final ATS evaluation report